Two-Regime Models for Daily Soil-Moisture Estimation in Washington

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Background and research question

Why soil moisture estimation?

Soil moisture regulates agricultural productivity and drought resilience, making daily estimates vital for water management. High-resolution modeling tracks moisture recharge and drydown dynamics, particularly across regions lacking direct measurement infrastructure.

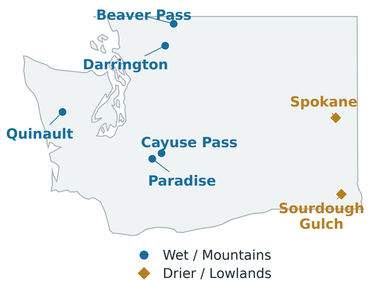
While in-situ ground sensors offer accurate readings, they exist only at limited monitoring sites. Satellite observations span broader regions but remain intermittent with frequent data gaps. To generate daily estimates, multi-source modeling combines satellite, weather, and site data, extending an established pipeline and global XGBoost baseline.

Summer research question

Can two specialists improve estimates across Washington's contrasting conditions?

Both models are compared under identical benchmarks. For known stations, we assign a fixed specialist; tests at new sites evaluate dynamic routing under missing satellite inputs.

Study area and data



Regional contrast

Stations span coastal areas, the Cascade mountains, and arid eastern Washington, capturing contrasting hydroclimatic regimes. Leave-one-station-out evaluations test model generalization across unseen microclimates.

Figure 1. Seven Washington training stations and their assigned clusters. Colors designate cluster regimes, with no geographic boundaries implied.

In-situ ground truth for soil moisture comes from NOAA USCRN and SNOTEL stations. Satellite inputs comprise SMAP soil moisture, Sentinel-1 radar backscatter from the land surface, and Sentinel-2 surface reflectance and indices of vegetation greenness and canopy water content. We use MODIS land-surface temperature, ERA5-Land weather, and terrain, soil, and climate descriptors.

The data pipeline aligns observations onto a daily grid, imputes missing values, smooths temporal sequences, and extracts rolling lag features capturing recent weather and satellite signals.

2017 to 2022
Final fitting period → **2023 to 2025**
Later-year test

We train the models with data from 2017 to 2022. We evaluate temporal generalization on later years (2023 to 2025) and test new Washington sites using leave-one-station-out cross-validation.

Two-regime modeling

A single global model enforces one input-to-moisture mapping across all stations. We divide observations into two environmental regimes and fit a dedicated XGBoost specialist to each. Each specialist predicts daily volumetric soil moisture at 5 cm depth (m^3/m^3).

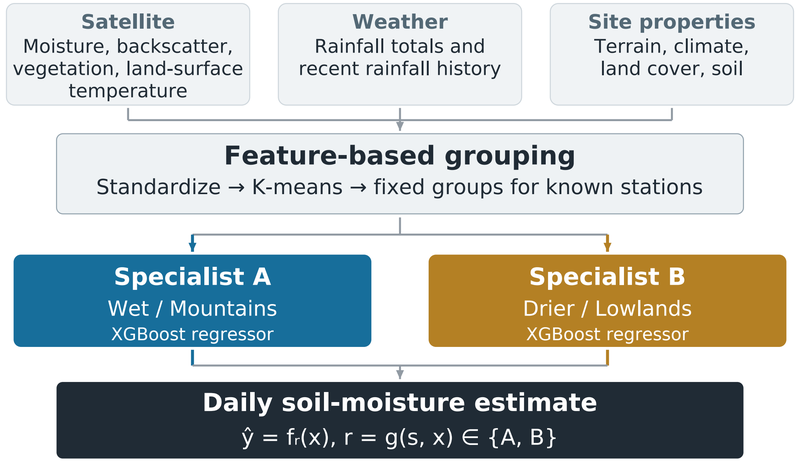


Figure 2. Two-regime routing: each observation is assigned to a specialist f_r, where x denotes input features, s the station, and g the routing function.

1. Router inputs are imputed using training-set means and standardized to comparable scales. K-means clustering identifies two centroids, assigning unseen samples to the nearest centroid using feature inputs alone without target labels.

2. For known stations, each adopts its majority cluster specialist for temporal consistency. New stations are routed dynamically each day, falling back to rainfall rules if satellite features are missing.

3. Both approaches use identical features, XGBoost settings, evaluation splits, and random seeds. Each specialist trains solely on its assigned regime. Testing 30 repeated seeds quantifies training stability, while station-holdout splits evaluate spatial generalization uncertainty.

Regime interpretation

Wet / Mountains

5 training stations

Western and Cascade sites: more rain and higher SMAP soil moisture.

Wettest-month rain: 391 mm
SMAP soil moisture: 0.414 m^3/m^3

Drier / Lowlands

2 training stations

Spokane and Sourdough Gulch: less rain and lower SMAP soil moisture.

Wettest-month rain: 70 mm
SMAP soil moisture: 0.192 m^3/m^3

The regimes separate primarily along gradients of rainfall and SMAP soil moisture. Because the wetter group includes lowland and mountain stations, elevation alone does not distinguish them. Displayed values are input medians; other deployments should select cluster counts based on local environmental data.

Evaluation and results

+0.0320
Later-year R^2 gain

+0.0581
Station-holdout R^2 gain

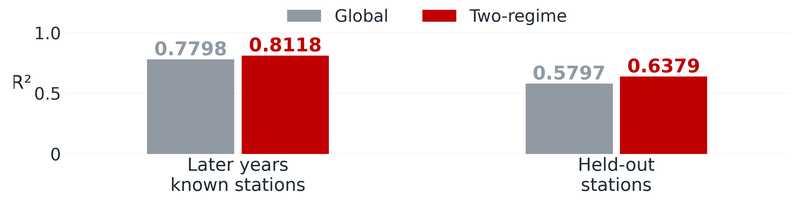


Figure 3. Mean R^2 across 30 later-year runs and five seeds per held-out station, using matched seeds (higher is better).

The two-regime model achieves higher R^2 across all 30 later-year runs. Spatial station holdouts also favor specialization on average (+0.0581 R^2), though gains vary by regional coverage.

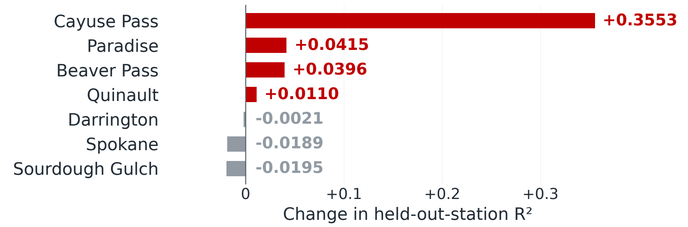


Figure 4. Held-out station R^2 differences (four gains and three losses).

Cayuse Pass gains +0.25 R^2. Conversely, withholding either eastern station removes half the drier group's data, underscoring sensitivity to sparse regional coverage.

ECE sensor-site evaluation

At five sensor sites deployed by Seattle University's ECE team, missing real-time SMAP retrievals triggered an automated fallback router using recent rainfall history. Regime thresholds were calibrated on Washington validation data without target labels.

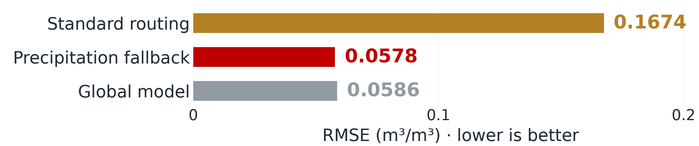


Figure 5. Pooled RMSE at ECE sites. The rainfall fallback matches global model error (lower is better).

During the dry summer period (July 20 to August 19, 2026; 150 daily observations), the fallback assigned all dates to the drier specialist. Full-season multi-regime performance remains untested.

Discussion and next steps

Specialist gains rely on accurate routing and representative training networks. Future work evaluates multi-season ECE sensor data and strengthens router robustness under intermittent satellite availability.